

Arad Fadaei

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SKILLS

Agentic AI: agent runtimes, native tool calling, human-in-the-loop guardrails, prompt/context design

Languages: Python, Javascript/Typescript, Rust, C/C++, Dart, Swift, Kotlin

Frameworks/Libraries: Svelte, React, NodeJS, NextJS, Express, Flutter

Tech: Git, Linux, VsCode, Vim, AndroidStudio, Xcode, Affinity Suite, MS Office

PROJECTS

Ethos - Personal AI Runtime [<https://github.com/Downmoto/ethos>]

Creator and lead developer | Python, asyncio, FastAPI, Pydantic, LiteLLM, Turso

- Architected an asynchronous agent runtime that streams reasoning and responses, executes validated tools, and supports OpenAI, Google, and Ollama through a provider-neutral contract.
- Built a FastAPI REST/SSE interface and CLI over one service boundary, with bearer authentication, durable sessions, workspace isolation, and interruption recovery.
- Designed safety controls for side-effecting tools: explicit human approval, persisted state transitions, replay protection, bounded results, cancellation, and indeterminate-write handling
- Implemented native process isolation on macOS and Linux with restricted filesystems, no network access, time and output limits, concurrent stream draining, and fail-closed behaviour.
- Added correlated lifecycle instrumentation for model runs and tool execution while excluding prompts, reasoning, credentials, and tool payloads from diagnostic events.
- Maintained strict static typing, architecture documentation, and a comprehensive automated test suite covering runtime, protocol, storage, capability, and failure-path behaviour.

aethellib - Rust Composable Text generation Library [<https://crates.io/crates/aethellib>]

Personal Open Source Project

- Engineered a composable text generation library in Rust, to load, process, and validate target-specific TOML corpora.
- Designed a programmatic generation engine utilizing modular rules (literals, picks, concatenation) instead of a rigid DSL, empowering binary developers to build predictable and extensible outputs.
- Implemented first-class source provenance tracking, ensuring generated values retain traceable metadata for clear data lineage and simplified debugging.
- Developed a context-aware state management system that pools values by section and field, allowing prior results to be stored and seamlessly reused across complex generation flows.
- Published the package to crates.io and maintained comprehensive documentation for user guidance on docs.rs.

PhishGuard - Machine Learning Phishing Detector [<https://github.com/Downmoto/nlp-PhishGuard>]

Machine Learning Project | Team Lead

- Fine-tuned a bert-base-uncased natural language processing model to detect phishing emails, achieving exceptionally high accuracy and robust predictive performance across a large, stratified test dataset.
- Engineered a robust data ingestion pipeline in Python that aggregates seven distinct datasets, cleans raw text (stripping HTML, masking URLs and PII), and exports partitioned Parquet files.
- Developed an interactive web interface using Gradio, featuring an attention-based explainability module that highlights the specific tokens driving the model's predictions.
- Constructed a fully automated, end-to-end command-line workflow that seamlessly connects data ingestion, training, and deployment scripts into a reproducible developer experience.

EDUCATION

Artificial Intelligence, Seneca College of Applied Arts and Technology

2026

Computer Programming and Analysis, Seneca College of Applied Arts and Technology

2025